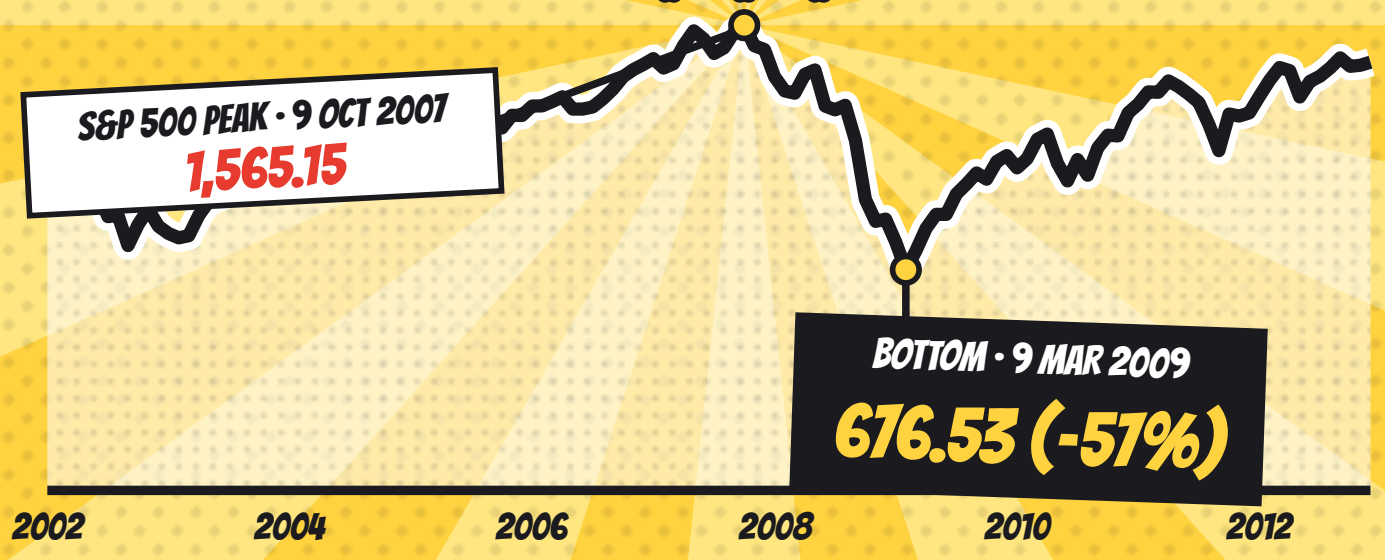
THE 2008 FINANCIAL CRISIS!

A HOUSE OF CARDS

2003 - 2009





House prices never fall. Everybody knows that!

Unless everybody borrowed too much to buy them...





A true story, told with real data · Chart: S&P 500, month-end closes (Yahoo Finance)

THE MORTGAGE MACHINE

What changed: home loans became something banks could sell. A new money machine turned family mortgages into bonds for investors all over the world.

1 A FAMILY BORROWS

A family takes out a mortgage: a loan to buy a home, paid back over about 30 years.

2 THE LENDER SELLS THE LOAN



The old way: the bank kept your loan, so it checked you could pay. The new way: sell the loan on within weeks. If it goes bad, it's someone else's problem.

3 WALL STREET BUNDLES IT

Investment banks pack thousands of mortgages into one bond, then slice it into layers. Families' monthly payments flow to the bond's owners.

4 RATED "AAA"



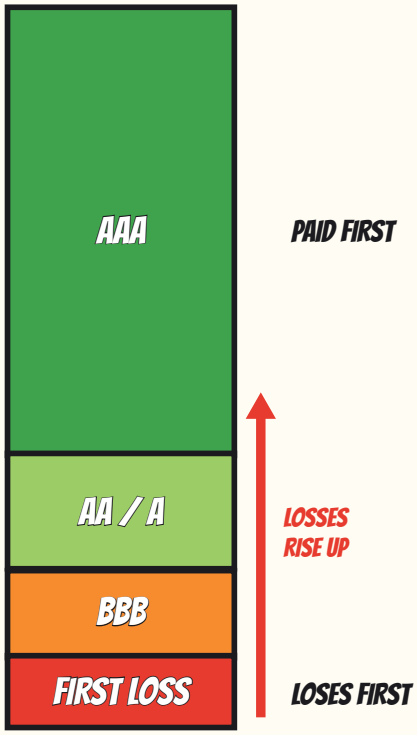
Rating agencies stamped most layers AAA: as safe as it gets. Their fees were paid by the banks selling the bonds.

5 SOLD WORLDWIDE



Pension funds, insurers and banks everywhere bought them, even eight small towns in Norway, which lost tens of millions of dollars.

ONE BOND, SLICED



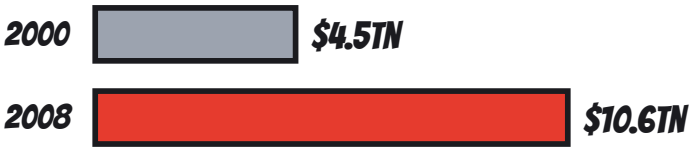
Payments flow in at the top. When families stop paying, the bottom layers lose first, so the top looked safe.

RISKY LOANS GREW



red = subprime: 1 in 20 loans, then 1 in 5

MORTGAGE DEBT DOUBLED



US home loans owed by households, trillions

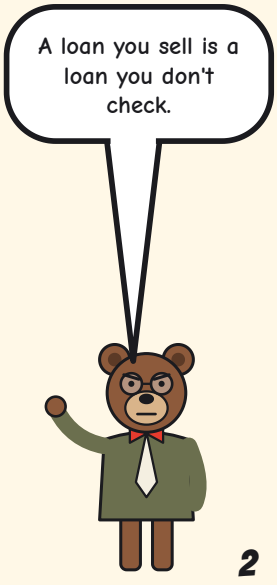
? WHAT DOES IT MEAN?

MORTGAGE
A loan to buy a home. If you stop paying, the lender can take the house.

MORTGAGE BOND
Thousands of mortgages bundled together and sold to investors (an "MBS").

SUBPRIME
A loan to someone with a weak credit record or low income: riskier, so it charges more.

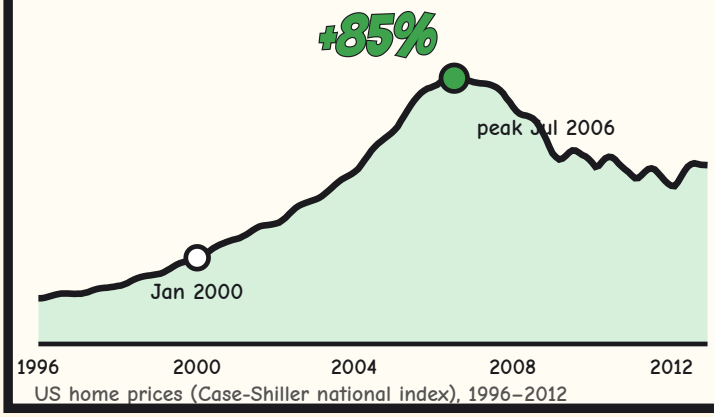
AAA
The top safety grade a rating agency can give. It means "almost no chance of loss".



THE GOOD TIMES

Before the crash, life felt good. These true facts made people believe that houses, jobs and shares could only go up.

HOME PRICES SOARED



US home prices (Case-Shiller national index), 1996–2012



69.2%

RECORD HOME OWNERSHIP

In 2004, 69.2% of US households owned their home, the highest ever recorded.



4.4%

PLENTY OF JOBS

Unemployment fell to 4.4% in late 2006 and stayed low into 2007.



$70.7TN

FAMILIES FELT RICH

US household wealth hit a record $70.7 trillion in mid-2007, much of it in rising house values.



CALM

"THE GREAT MODERATION"

Since the mid-1980s, growth and inflation had been unusually steady. Fed governor Ben Bernanke gave it that name in 2004.



14,164

SHARES HIT A RECORD

The Dow Jones index closed at a record 14,164.53 on 9 October 2007.



Buy a house now, before prices go even higher!

Prices were rising because loans were easy, and loans were easy because prices were rising.



FROM BOOM TO DANGER

Cheap money and easy loans pushed prices up. Underneath, the whole system was built on borrowed money.

2003

MONEY ON SALE



The Fed cut its rate to 1%, the lowest since 1958. 30-year mortgage rates fell to 5.2%, the lowest on record.

1

2004-06

LOANS FOR ALMOST ANYONE



Some lenders stopped checking incomes at all. Bankers nicknamed these "liar loans".

2

TEASERS

CHEAP NOW, DEAR LATER



Many subprime loans started with a low "teaser" rate for 2 years, then jumped. Borrowers hoped to refinance before then.

3

2005

BUYING TO FLIP



28% of all US homes bought in 2005 were investments, not homes to live in. Many buyers planned to sell quickly at a profit.

4

LEHMAN, END OF 2007



Lose $1 in every $31 and its own money is gone.

BANKS BORROWED TO THE HILT

At the end of 2007, investment bank Lehman Brothers held $691 billion of assets on only $22.5 billion of its own money: 30.7 times over. A fall of just over 3% in what it owned would wipe out all of its own money.

5

AIG, 2007



INSURANCE WITH NO MONEY BEHIND IT

AIG, a giant US insurer, had sold $513 billion of "credit default swaps": insurance that bonds would be repaid. It set little money aside to pay claims. When the claims came in 2008, it needed an $85 billion rescue from the Fed.



6

WHAT DOES IT MEAN?



LEVERAGE
Using borrowed money to bet bigger. Gains grow faster, but so do losses.



CREDIT DEFAULT SWAP
An insurance policy that pays out if a borrower can't repay a bond.



So if house prices fell just a little...?

...the whole borrowed tower could topple.

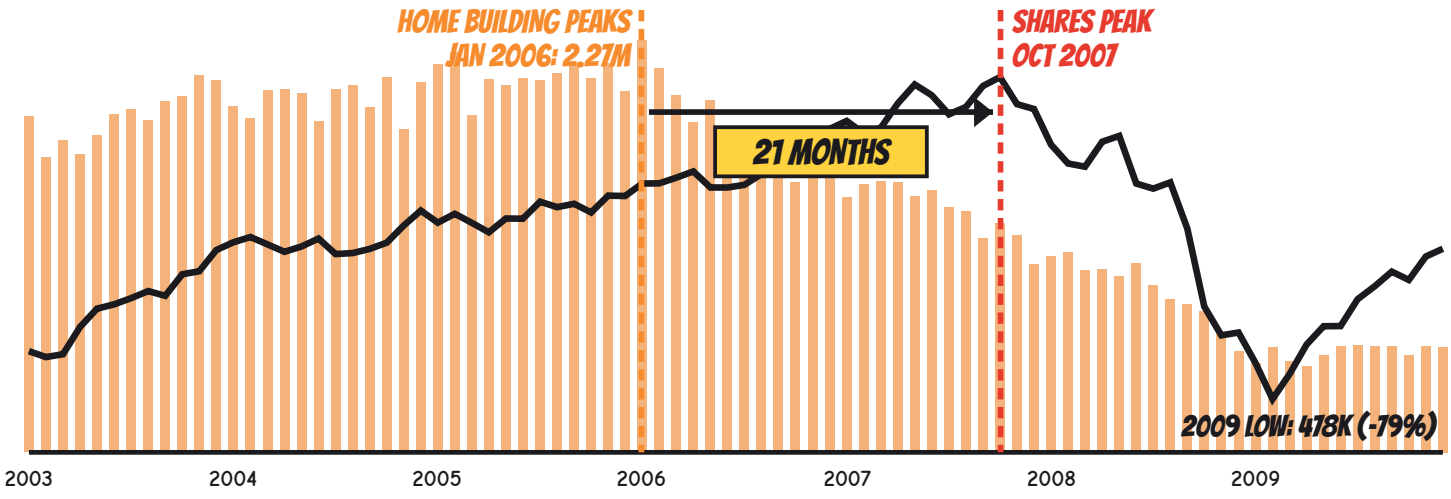


THE CRACKS SHOWED FIRST

Shares kept rising until October 2007. But two warning signs had been flashing for more than a year before that.

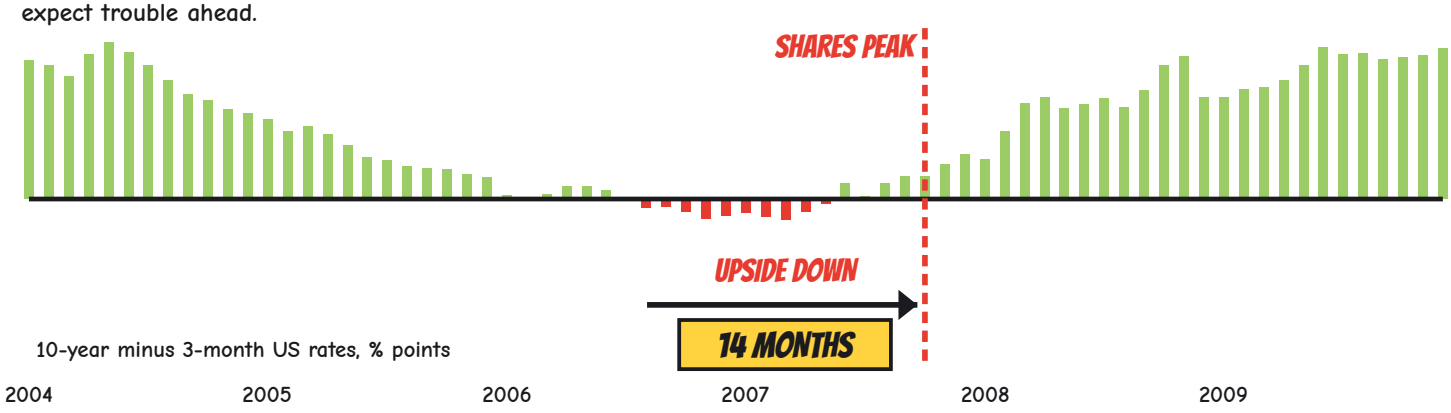
1. AMERICA STOPPED BUILDING HOMES

Bars: new homes started each month (thousands, yearly rate). Line: the S&P 500 share index.



2. THE YIELD CURVE TURNED UPSIDE DOWN

Normally, lending money for 10 years pays more than lending for 3 months. When it pays LESS (red), investors expect trouble ahead.



WHAT DOES IT MEAN?

HOUSING STARTS
How many new homes builders begin each month. Builders stop when buyers disappear.

YIELD CURVE
Interest rates for short and long loans, side by side. Upside down = a warning.

So the warning was in HOUSES, not in shares!

Both are lights on our scoreboard. They switched on 12 and 14 months before the peak.





TIMELINE PART 1: UP, UP...

Read from the bottom up. Green = good news, red = trouble, blue = a warning or a rescue. Red stamp = the crash.

LATER

9 OCT 2007 — CRASH STARTS

THE PEAK

S&P 500 closes at 1,565.15; the Dow at a record 14,164.53.



18 SEP 2007

FED STARTS CUTTING

Rates cut from 5.25% to 4.75%. Shares rally.



14 SEP 2007

RUN ON NORTHERN ROCK

A British lender needs emergency help. Savers queue to take their money out.



9 AUG 2007

BNP PARIBAS FREEZES FUNDS

A French bank can't value its US mortgage bonds: "a complete evaporation of liquidity".



2 APR 2007

NEW CENTURY GOES BUST

A top subprime lender fails as borrowers stop paying.



JUL 2006

HOME PRICES PEAK

After rising 85% since 2000, US home prices start to slide.



30 JUN 2004

FED STARTS RAISING RATES

17 rises in a row take rates from 1% to 5.25% by June 2006.



2004

HOME OWNERSHIP RECORD

69.2% of US households own their home.



25 JUN 2003

RATES CUT TO 1%

The lowest since 1958. Borrowing to buy a home has never been cheaper.



EARLIER

PART 2: ...AND DOWN!

Read from the bottom up. The crash: 9 Oct 2007 to 9 Mar 2009 (red stamps). The worst weeks came after Lehman fell.

LATER

9 MAR 2009 — CRASH ENDS

THE BOTTOM

S&P 500 676.53 (-57%). Dow 6,547.05 (-54%).



16 DEC 2008

RATES CUT TO ZERO

The Fed's rate goes to 0–0.25%. Recession: Dec 2007–Jun 2009.



6-9 OCT 2008

ICELAND'S BANKS COLLAPSE

All three big banks fail; together they were about 10 times Iceland's economy.



3 OCT 2008

RESCUE LAW SIGNED

A revised $700 billion rescue fund (TARP) becomes law.



29 SEP 2008

CONGRESS SAYS NO

The House rejects a $700 billion bank rescue: S&P 500 -8.8% in a day.



16 SEP 2008

AIG RESCUED

The Fed lends $85 billion and takes 79.9% of the insurer.



15 SEP 2008 — THE CRASH

LEHMAN BROTHERS FAILS

$639 billion of assets: the biggest bankruptcy in US history. S&P 500 -4.7%.



7 SEP 2008

FANNIE & FREDDIE TAKEN OVER

The government seizes the two giants behind half of US mortgages.



16 MAR 2008

BEAR STEARNS SOLD FOR $2

A Wall Street bank is sold to JPMorgan for $2 a share (later $10), backed by the Fed.



11 JAN 2008

COUNTRYWIDE RESCUED

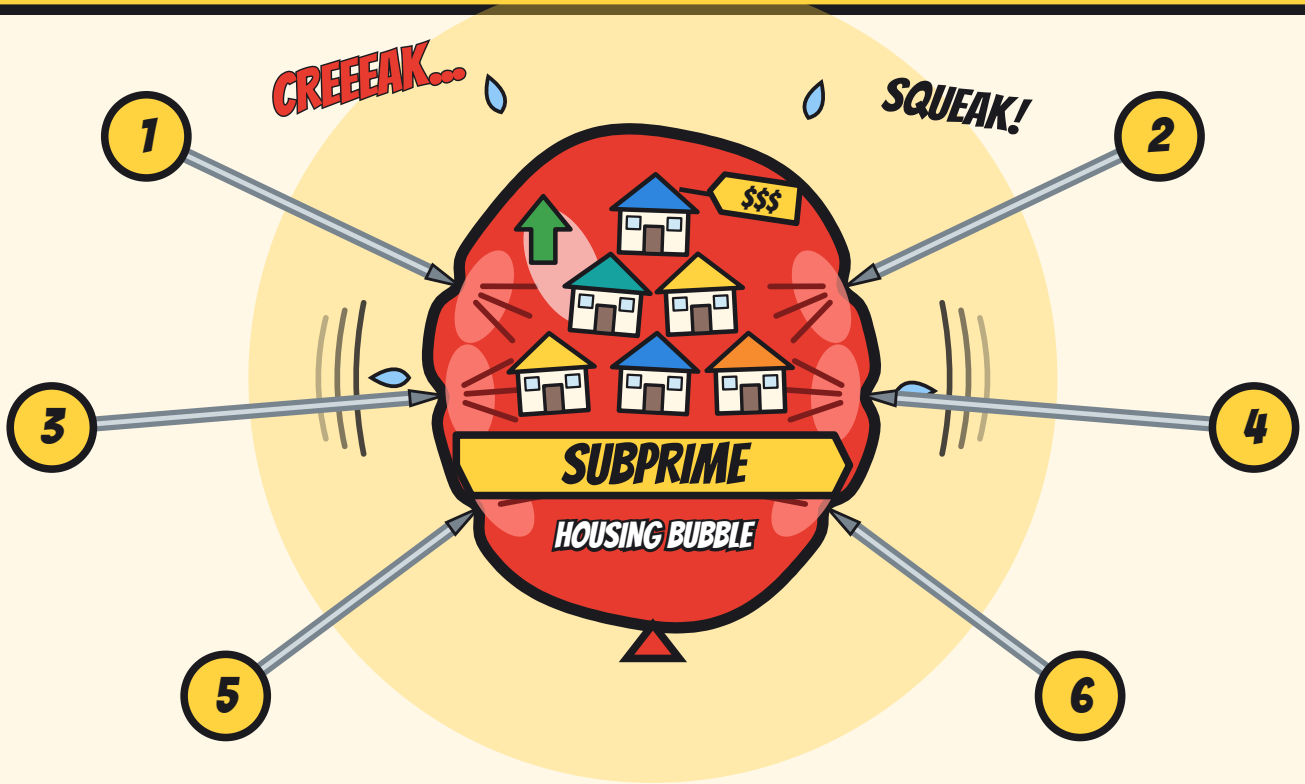
Bank of America buys the biggest US mortgage lender for about $4 billion.



EARLIER

WHAT POPPED IT?

Six pins pressed into the housing bubble. Each made the next one worse:



1 DEARER MONEY

The Fed raised rates 17 times, from 1% to 5.25% (2004–06). Loan payments rose.



2 TEASERS RAN OUT

Cheap starter rates reset higher. Monthly payments jumped for millions of families.



3 PRICES FELL

Home prices fell 27% from July 2006 to early 2012. Many owed more than their home was worth.



4 PEOPLE STOPPED PAYING

Late mortgage payments rose from 1.4% (2004) to 11.5% (2010) of home loans at banks.



5 BORROWED MONEY

Banks borrowing 30 times their own money could not survive even small losses.



6 TRUST FROZE

Nobody knew which bank held the bad loans, so banks stopped lending to each other.

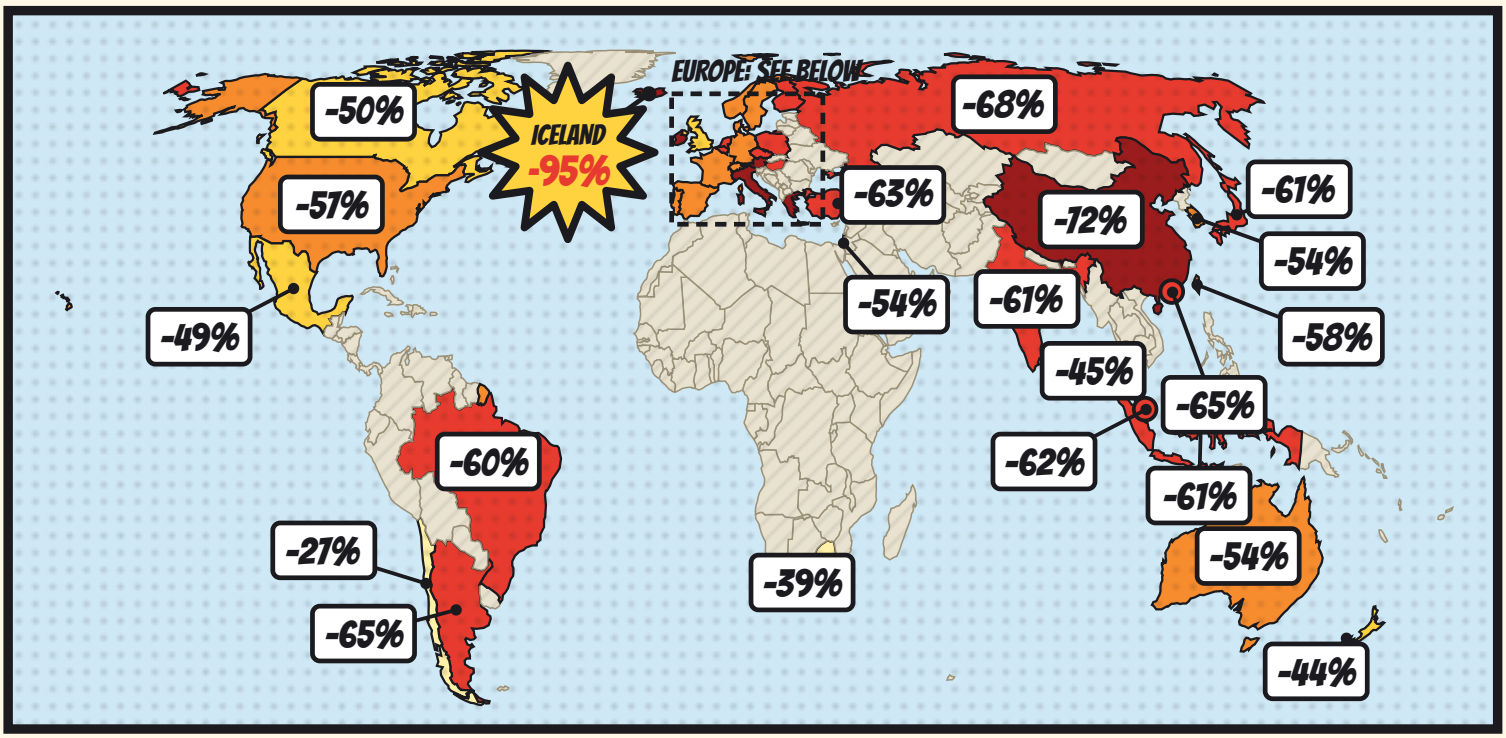


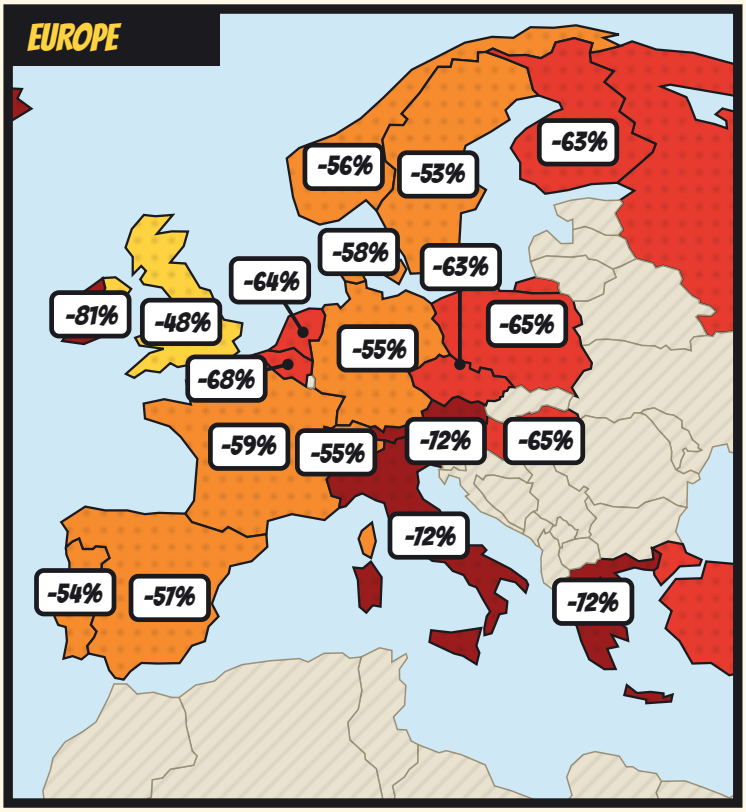


Our warning lights: home building and the yield curve flashed a year or more before the stock peak. Credit spreads only switched on one month before.

WHO GOT HURT?

Bad US mortgages were owned all over the world, so the crash went global. Each country is shaded by how far its main stock index fell from its 2007–08 high to its low by the end of 2009.





THE FALLS, RANKED

Iceland‡	-95%	France	-59%
Ireland	-81%	Denmark‡	-58%
Greece	-72%	Taiwan	-58%
China†	-72%	Spain	-57%
Austria	-72%	United States	-57%
Italy	-72%	Norway‡	-56%
Russia‡	-68%	Germany	-55%
Belgium	-68%	Switzerland	-55%
Hong Kong	-65%	South Korea	-54%
Poland‡	-65%	Israel	-54%
Hungary‡	-65%	Australia	-54%
Argentina	-65%	Portugal‡	-54%
Netherlands	-64%	Sweden‡	-53%
Turkey	-63%	Canada	-50%
Finland‡	-63%	Mexico	-49%
Czechia‡	-63%	United Kingdom	-48%
Singapore	-62%	Malaysia	-45%
Japan	-61%	New Zealand	-44%
India	-61%	South Africa‡	-39%
Indonesia	-61%	Chile‡	-27%
Brazil	-60%		

INDEX FALL: 70% or more | 60–70% | 50–60% | 40–50% | less than 40% | no data used



Banks everywhere owned US mortgage bonds.

Iceland fell 95%?! How?



Its three banks had grown to about 10 times the size of its whole economy. When they failed, so did its market.

Local-currency index closes, Yahoo Finance; ‡ = OECD monthly average share prices (FRED), which hide the worst days. Map: Natural Earth.
† China's market had its own bubble in 2006–07; it peaked on 16 Oct 2007 and fell 72%.

FIXING THE RULES

Governments spent the next ten years rewriting the rules so that banks hold more of their own money and taxpayers do not have to rescue them again.



21 JUL 2010

UNITED STATES

DODD-FRANK ACT

Yearly stress tests for big banks, a council to spot system-wide risks, and a way to close a failing giant without a bailout.



21 JUL 2011

UNITED STATES

CONSUMER FINANCIAL PROTECTION BUREAU

A new watchdog for mortgages, credit cards and other loans to families.



10 JAN 2014

UNITED STATES

"ABILITY TO REPAY" RULE

Lenders must check that a borrower can afford the loan. No more "liar loans".

BASEL

16 DEC 2010

WORLDWIDE

BASEL III BANK RULES

Banks must hold more of their own money, keep cash for emergencies and limit how much they borrow.

G20

25 SEP 2009

G20 COUNTRIES

DERIVATIVES REFORM

Swaps like AIG's must go through central clearing houses that hold money to cover losses.

FSB

2 APR 2009

WORLDWIDE

FINANCIAL STABILITY BOARD

Set up by the G20 to watch for risks across the whole world's financial system.



18 DEC 2013

UNITED KINGDOM

RING-FENCING (BANKING REFORM ACT)

Big banks must wall off everyday savings from riskier investment banking, fully from 1 Jan 2019.



4 NOV 2014

EURO AREA

ECB BANK SUPERVISION

The European Central Bank checks the biggest eurozone banks, instead of 19 separate national watchdogs.



15 MAY 2014

EUROPEAN UNION

"BAIL-IN" RULES (BRRD)

If a bank fails, its owners and lenders take the losses first, not taxpayers.

Banks now must hold far more of their own money and pass a yearly "stress test". Their rescues are now planned in advance, so that their owners lose first, not taxpayers.



So could it happen again?



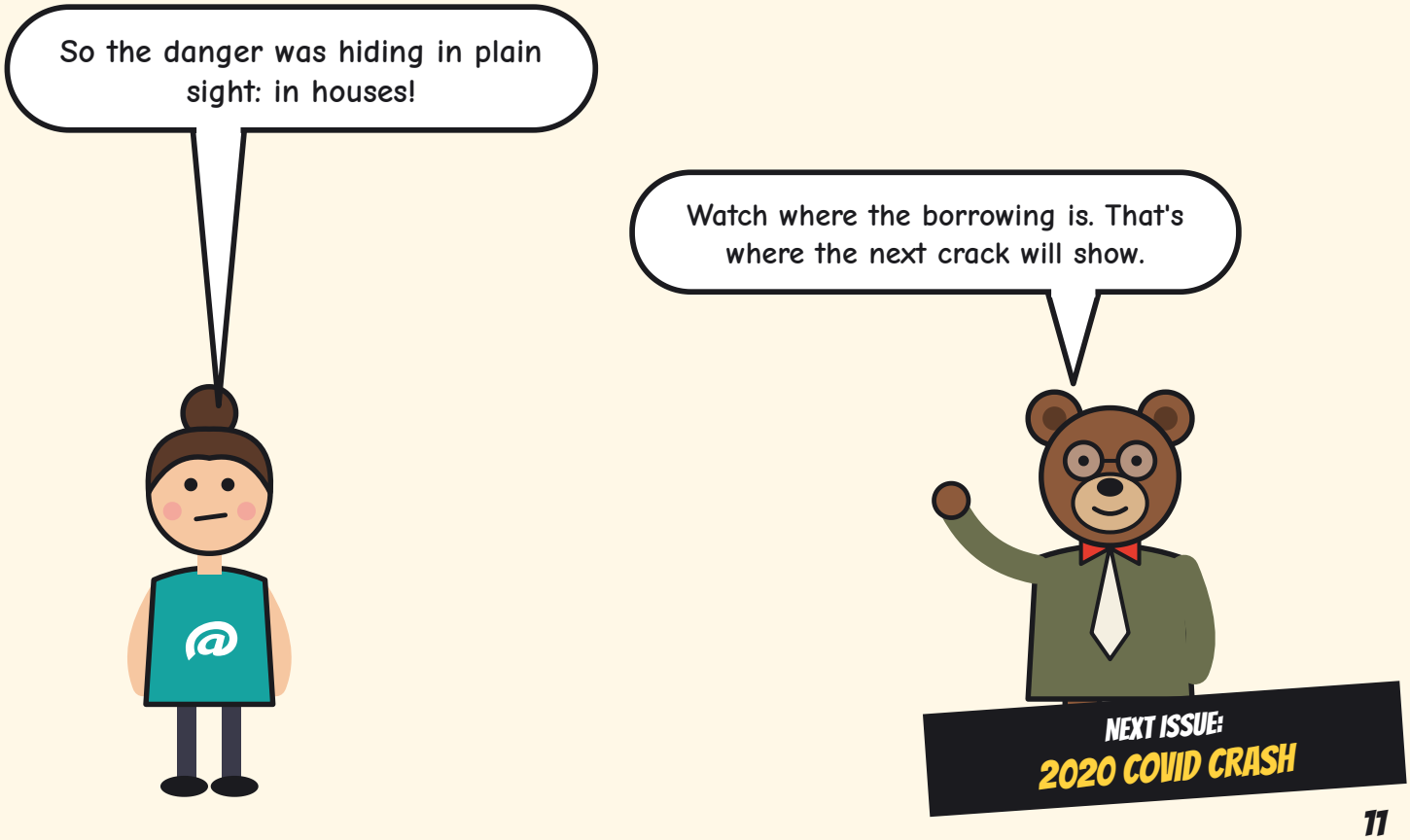
WHAT OUR WARNING LIGHTS SAID

The same 10 rules we use for every crash, applied to the months before the October 2007 peak.

YIELD CURVE INVERTED WARNED 14 months early	**FED HIKING FAST** LONG ON on for 31 months	**INFLATION HIGH** TOO LATE 1 month after	**OIL SHOCK** TOO LATE 1 month after	**CREDIT SPREAD WIDENING** WARNED 1 month early
CORPORATE DEBT BUILD-UP TOO LATE 2 months after	**STOCKS EXPENSIVE** WARNED 23 months early	**PROFITS FALLING** WARNED 7 months early	**HOUSING SLUMP** WARNED 12 months early	**FACTORY HOURS CUT** WARNED 8 months early

6 of 10 lights warned early. Home building fell for a year before the peak and the yield curve flipped 14 months ahead; stocks were expensive and profits were falling. Credit spreads, the bond market's fear gauge, only switched on one month before.

THE LESSON: Housing gave the clearest warning — the stress showed up where the borrowing was. Home building had collapsed and the yield curve had inverted more than a year before the peak.



NEXT ISSUE: 2020 COVID CRASH

SOURCES & NOTES

Every number in this comic was checked against the sources below on 21 October 2026. Index falls use the highest close from January 2007 to June 2008 and the lowest close after it up to December 2009, in local currency. US home prices: S&P CoreLogic Case-Shiller national index. Jobs, rates, debt and wealth: official US data (BLS, Census, Federal Reserve) via FRED. Characters are fictional. For education only, not investment advice.

THE MORTGAGE MACHINE (PAGE 2)

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THE GOOD TIMES (PAGE 3)

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FROM BOOM TO DANGER (PAGE 4)

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WARNING SIGNS AND WHAT POPPED IT (PAGES 5, 8)

https://fred.stlouisfed.org/series/HOUST
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FIXING THE RULES (PAGE 10)

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TIMELINE AND DATA

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https://www.federalreservehistory.org/essays/fannie-mae-and-freddie-mac (Fannie & Freddie)
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https://finance.yahoo.com/ (daily index closes) · https://www.naturalearthdata.com/ (map outlines)